Context-Aware Semantic Construction for Multimodal Outfit Recommendation: A Data-Driven Framework with Thermal-Comfort Proxy Variables and LLM-Based Generation-Quality Diagnostics

Context-Aware Semantic Construction for Outfit Recommendation

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Context-aware outfit recommendation requires more than visual compatibility: outfits should also reflect weather, activity, occasion, and style. We frame this data-centric multimodal recommendation problem by constructing traceable natural-language conditions from outfit images and text. The pipeline estimates outfit-level clothing insulation, maps plausible activities to Metabolic Equivalent of Task values, derives a temperature reference through simplified thermal mapping, and rewrites short Polyvore descriptions into context-aware queries. CLO and MET values are derived from established thermal-comfort references and are used as proxy variables rather than measured comfort labels. Generation quality is diagnosed with checklist-based LLM-as-a-Judge and a stratified manual audit, while recommendation performance is evaluated with a fixed Hybrid Attention text-conditioned recommender. On 35,140 Polyvore Outfits Disjoint samples, context-aware descriptions improve compatibility and conditional item retrieval: AUC increases from 0.9292 to 0.9454, FITB accuracy from 0.6374 to 0.6478, Recall@10 from 0.0780 to 0.0921, and Recall@50 from 0.2303 to 0.2518. Semantic ablations show style contributes most consistently, while weather and occasion add complementary constraints. Simpler Two-Tower validation preserves retrieval gains, and controlled counterfactual replacements alter rankings observably. These results indicate that traceable context-aware semantic construction can improve multimodal outfit retrieval without modifying the primary recommender architecture, while LLM-based generation-quality diagnostics remain supportive checks rather than human ground truth.

CCS CONCEPTS • Information systems → Recommender systems; Multimedia information systems; Retrieval models and ranking • Computing methodologies → Artificial intelligence; Natural language processing; Computer vision

Additional Keywords and Phrases: multimodal recommendation, outfit recommendation, context-aware semantic construction, LLM-as-a-Judge

1 INTRODUCTION

For fashion recommender systems to move beyond commerce-oriented product recommendation toward decision support for everyday dressing, they must address contextual factors that are more complex than product preference and visual compatibility alone. Prior fashion recommendation research has progressed from item-level preference modeling to multimodal compatibility, image-text retrieval, and generative outfit construction, yet most methods still focus primarily on relationships among products or items and less often address constraints that users encounter in real-life dressing situations. Everyday dressing is not merely a matter of selecting individual items. It is a compound decision influenced by weather, planned activities, usage scenarios, occasion formality, and overall styling requirements. Consequently, an outfit that appears visually coherent may still be unsuitable in practice if it does not match the current setting, climate, or style requirements.

This limitation also affects multimodal recommendation data. Even when a model can jointly use image and text information, if the data lack contextual conditions that the model can learn from, the model may primarily capture visual compatibility among items and fail to respond consistently to users' actual weather, occasion, and style requirements. In

other words, whether a recommendation model can respond effectively to outfit context depends not only on the model architecture but also on whether the data provide sufficiently explicit, traceable, and auditable contextual semantics.

Accordingly, we formulate context-aware outfit recommendation as a data-centric multimodal recommendation problem. Because mood, personal preference, and real usage context are difficult to recover comprehensively from existing datasets, we focus on three contextual dimensions that can be explicitly constructed for model learning: weather, occasion, and style. Using clothing insulation (CLO), Metabolic Equivalent of Task (MET), simplified thermal-comfort estimation, and large language model (LLM)-assisted rewriting, we transform short outfit descriptions that lack contextual information into text conditions enriched with contextual semantics. A fixed Hybrid Attention text-conditioned outfit recommendation model serves as the primary controlled experiment for examining the effect of data-side context-aware semantic construction on offline recommendation performance, while a simpler Two-Tower architecture is used for cross-architecture robustness validation. Rather than introduce a new recommendation backbone, we show how data-side context-aware semantic construction addresses original outfit-data limitations and affects recommendation performance, contextual-factor behavior, and candidate-ranking patterns. In the main comparison, "held fixed" means that the data split, FashionCLIP representations, Hybrid Attention architecture, hyperparameters, and training procedure remain identical across text conditions; only the input text source changes.

1.1 Research Questions

This study examines data-side context-aware semantic construction through four research questions:

- RQ1: Do context-aware descriptions improve outfit compatibility and conditional item retrieval when the recommendation model is held fixed?
- RQ2: How do weather, occasion, and style semantics affect outfit compatibility and rankings?
- RQ3: Can clothing insulation, activity MET, and temperature reference values provide a traceable and reasonable basis for constructing contextual conditions?
- RQ4: Are the generated descriptions sufficiently image-text consistent, contextually plausible, and auditable to serve as reliable text conditions for downstream recommendation evaluation?

1.2 Contributions

- We propose a data-centric context-aware semantic construction pipeline for multimodal outfit recommendation integrating outfit images, original descriptions, CLO/MET proxies, and simplified thermal mapping into natural-language conditions learnable by a recommendation model.
- We use interpretable CLO and MET scales derived from established thermal-comfort references to constrain weather and activity semantics; they serve as proxy variables rather than measured comfort labels.
- We use fixed Hybrid Attention with FashionCLIP representations so the main comparison isolates the text source. A simpler FashionCLIP Two-Tower model provides cross-architecture validation.
- We design a multi-level evaluation covering recommendation performance, semantic sensitivity, and diagnostic reliability, combining the main recommendation tasks, semantic ablations, cross-architecture and counterfactual robustness analyses, generation-quality diagnostics, and manual auditing to characterize the effects, mechanisms, and limits of context-aware descriptions.

2 RELATED WORK AND RESEARCH POSITIONING

This section reviews multimodal and text-conditioned outfit recommendation, LLM-assisted data construction, and contextual fashion recommendation, then positions the present study against the resulting data-quality and evaluation gaps.

2.1 Multimodal Outfit Recommendation and Text-Conditioned Retrieval

Fashion recommender systems were initially developed largely for e-commerce, where the primary objective is to help users search, rank, and filter apparel items from large catalogs. Because online fashion catalogs are large and rapidly changing while user interaction records are relatively sparse, prior studies have commonly addressed personalized product recommendation, similar-item recommendation, or complementary-item recommendation using interaction histories, product attributes, visual features, or textual descriptions to predict individual items or matching products that a user may prefer. These approaches support product discovery and e-commerce conversion, but when applied to everyday dressing decisions they may still be insufficient for determining whether an entire outfit fits the user's current life context[7, 8].

Unlike item-level product recommendation, outfit recommendation is a compositional decision problem. It must determine not only whether an individual product is worth recommending or whether a candidate is similar or compatible with an existing item, but also whether multiple items from different categories can form a wearable, appropriate, and coherent outfit. Fashion-recommendation reviews note that outfits span multiple categories and, beyond personal preference, require item compatibility and overall coherence. Nevertheless, existing datasets and model settings still tend to emphasize product relationships, visual compatibility, or candidate-item ranking, with less direct treatment of real-world constraints such as warm versus cold weather, planned activities, occasion formality, and style requirements.

Recent work has incorporated natural-language descriptions into outfit recommendation conditions so that models can respond not only to product images or category information but also to dressing requirements expressed in text. Wang and Zhong proposed Text-Conditioned Outfit Recommendation with a Hybrid Attention Layer that jointly models textual conditions and item features to construct outfits that satisfy the description while maintaining internal compatibility [28]. Jang et al. introduced the Text-to-Outfit Retrieval task, which maps a user's textual description directly to a complete outfit and models text-outfit relationships at the item, style, and outfit semantic levels [16]. Recent studies further show that large multimodal models can extract fashion attributes for outfit-compatibility prediction, while style-identifiable multimodal models explicitly couple compatibility with style representation [4, 24].

These text-conditioned approaches, however, remain highly dependent on the quality of the original textual conditions in the data. If outfit descriptions are very short or contain only product names, thematic tags, and style phrases, a model may learn image-text alignment and item compatibility without learning the weather, occasion, and activity constraints needed for everyday dressing. We therefore shift the research question to the data side rather than simply applying existing product recommendation methods to an outfit task: how can commerce-oriented outfit data with limited life-context information be transformed into traceable and auditable context-aware text conditions that a fixed recommendation model can learn?

2.2 LLM-Assisted Recommendation and Data-Centric Challenges

Large language models (LLMs) are widely studied in recommender systems and support user-intent understanding, text representation, feature engineering, product-description enrichment, preference elicitation, and explanation generation. Their value to recommendation lies not only in directly producing recommendations, but also in transforming sparse or incomplete text into semantically richer auxiliary features that can improve user or item representations [19, 29].

Using LLMs to generate recommendation inputs, rank candidates, or recommend items also introduces limitations. Prior work notes that LLMs often incur substantial inference cost and latency and may generate plausible but unsupported content. If such erroneous attributes or descriptions are incorporated into recommendation models, they may affect both model decisions and explanations [19]. Likewise, when LLMs are used for automated evaluation, their judgments may depend on prompt design, model bias, evaluation criteria, and calibration procedures, motivating standardized procedures, additional validation, and robustness checks [13].

Given these limitations, we adopt a data-centric perspective in which the LLM is not the final recommender but a tool for constructing and diagnosing textual conditions. Specifically, LLMs are used to rewrite context-aware descriptions and assess generation quality, while fixed prompts, structured outputs, checklist rules, and sample-level audits are used to reduce the effects of unstable generation and hallucinated attributes on downstream training. The main recommendation experiments hold the recommendation backbone, data split, and training settings constant so that the core comparison isolates whether the input text provides more useful contextual semantics. Additional recommendation architectures are used only for complementary robustness validation of whether the observed gains depend strongly on a particular model design.

2.3 Context-Aware Fashion Recommendation and Thermal-Comfort Proxy Variables

Context-aware recommendation assumes that relevance changes with the usage context rather than being determined solely by fixed user preferences, historical interactions, or item similarity. In fashion recommendation, reviews have identified location, season, occasion, culture, and social background as factors that may affect whether a recommendation meets practical dressing needs. Weather, special occasions, location, and scene information have likewise been highlighted as important directions for making fashion recommendation more specific and practically useful [7, 8]. These conditions are often handled separately, however, through weather-aware, occasion-aware, or location-aware recommendation rather than being converted into integrated natural-language conditions that a text-conditioned outfit recommender can learn.

For everyday dressing, we define context-aware outfit recommendation as augmenting item compatibility and styling coherence with information that influences clothing choices, represented as model-learnable text conditions. We operationalize context along three dimensions: weather, occasion, and style. Weather captures external climate and thermal constraints, occasion captures activity and social-context constraints, and style captures the overall visual semantics and aesthetic direction of the outfit. This definition does not restrict context awareness to the physical environment, but includes dressing conditions that users commonly express together in natural-language queries.

Weather refers to external climatic and perceived thermal conditions that affect comfort and clothing choice, including temperature, season, thermal sensation, precipitation, and insulation needs. Prior weather-oriented fashion recommendation work has incorporated weather as an input condition, demonstrating its relevance to clothing recommendation [20]. Research on outdoor clothing choice also shows that air temperature, humidity, and wind speed may affect clothing insulation and garment selection, while consumer research suggests that weather can influence color preferences [6, 17]. In this study, weather therefore represents comfort, insulation needs, and external environmental constraints rather than serving merely as background information.

Occasion covers dressing purpose, social context, formality, time, and setting. It is not limited to geographic locations such as an office, wedding venue, or travel destination, but also covers commuting, dating, gatherings, and evening activities that combine purpose and temporal context. Prior occasion-aware outfit research shows that outfit appropriateness depends not only on visual coordination among items but also on context and occasion requirements [27]. Accordingly, occasion in this study is not a single location label. It describes the activity and social context in which an

outfit is worn. A standalone temporal expression such as “evening” is treated as a temporal modifier within occasion semantics, while a more complete occasion is formed when it co-occurs with activity cues such as a party, date, dinner, or formal event.

Style denotes the overall styling semantics formed jointly by visual cues such as garment silhouette, material, color, pattern, item combination, and accessories. Unlike weather and occasion, style is not an external physical condition and is not equivalent to a user's personal taste. Rather, it describes the aesthetic direction and visual tone conveyed by an outfit. Fashion-style research indicates that style can be composed from garment types, colors, materials, details, and multiple style attributes and therefore should not be reduced to a single fixed label [1]. Reviews of fashion recommendation likewise note that aesthetic factors affect clothing choices and may not be fully captured by conventional visual features [18]. We therefore include style as a contextual condition to represent overall outfit semantics not captured by weather or occasion.

To constrain free-form text generation, we introduce CLO and MET as thermal-comfort proxy variables in the data-construction process. CLO represents the thermal insulation of clothing, whereas MET represents metabolic intensity. Both are important parameters in human heat exchange and thermal-comfort reasoning[11, 15]. We then use McIntyre's simplified mapping derived from Fanger's thermal-comfort model to convert CLO and MET into a temperature reference value that constrains weather-semantic generation and provides a more traceable connection among clothing insulation, activity intensity, and textual description[21]. Recent computer-vision research has also estimated CLO and MET dynamically, reinforcing their continued role as interpretable personal parameters in thermal-comfort modeling [25].

The CLO, MET, and temperature reference values used in this study are data-construction proxies rather than measurements from human-subject experiments or true comfort labels. We do not claim to predict an individual's actual thermal comfort. Instead, these physical reference quantities provide a more consistent and traceable basis for linking outfit insulation, activity intensity, and weather semantics. The scope of this work is therefore context-aware semantic construction under physically motivated proxy constraints, together with offline utility evaluation using a fixed primary recommendation model and complementary robustness validation.

2.4 Research Gap and Positioning

The preceding literature reveals three issues. First, multimodal outfit recommendation can already model outfit compatibility from images, text, and item relations, and recent text-conditioned approaches further incorporate natural-language descriptions. However, when outfit text consists mainly of product names, style tags, or short thematic phrases, models may learn image-text alignment and item relationships without reliably capturing everyday semantics such as weather, activity intensity, occasion formality, and insulation needs. Thus, outfit recommendation performance depends not only on model architecture but also on whether the input text provides sufficiently explicit and traceable contextual information[7, 16, 28].

Second, LLMs can support semantic enrichment and data construction in recommender systems, but without explicit constraints and quality control, generated descriptions may introduce visually unsupported content, semantic drift, or unstable outputs. Prior work similarly notes that LLM-assisted recommendation must address inference cost, hallucination, and reliability, while LLM-based evaluation requires standardized procedures, calibration, and robustness checks [13, 19, 29]. When LLMs are used to rewrite outfit descriptions, the central issue is not text volume but whether generated content is constrained, auditable, and linked to downstream performance under a fair comparison.

Third, context-aware fashion recommendation has recognized the influence of weather, season, location, and occasion, but these factors are often incorporated as external conditions, auxiliary fields, or task-specific variables rather than being transformed into integrated natural-language conditions for text-conditioned outfit recommendation. Everyday dressing

needs are rarely expressed as a single label. Instead, weather, occasion, and style jointly form the semantic context. It is therefore important to investigate how these factors can be converted into traceable and auditable text conditions that can train a fixed multimodal recommendation model[1, 7, 20, 27].

Based on this positioning, we treat context-aware description construction itself as a data-centric multimodal outfit recommendation problem. We do not propose a new recommendation backbone. Instead, a fixed text-conditioned outfit recommender is used as the primary controlled setting to compare original outfit descriptions and context-aware descriptions under the same data split and training conditions, with a simpler architecture providing cross-architecture robustness validation. Specifically, clothing insulation, MET, and temperature reference values provide proxy constraints for LLM-based description rewriting. We then combine generation-quality diagnostics and manual auditing, text-length and target-clue checks, semantic-factor ablation, cross-architecture validation, and counterfactual sensitivity analysis to evaluate the contextual text conditions from the perspectives of data quality, recommendation utility, and ranking behavior. The central contribution is not to demonstrate the superiority of an LLM or a particular recommender architecture, but to examine how traceable and auditable contextual text conditions affect multimodal outfit recommendation and how robust those effects are across analyses and model settings.

3 METHODOLOGY

This section defines the data-construction problem and study scope, then describes context-aware description construction, generation-quality diagnostics, and the downstream recommendation setup.

3.1 Problem Formulation and Study Overview

Before detailing the individual components, this subsection formalizes the data-transformation objective, dataset scope, data-centric motivation, and downstream recommendation evaluation.

3.1.1 Problem Definition

Let an outfit be denoted by $O = i_1, i_2, \ldots, i_n$, where each item ij contains information such as an image, category, and textual attributes. The original dataset provides an outfit-level description d0, which is typically short and primarily reflects product names, style tags, or brief thematic phrases. Our goal is to construct a context-aware description de without changing outfit composition or semantics while adding weather, occasion, and style information relevant to everyday dressing.

Specifically, we formulate context-aware description construction as a data-transformation problem. The construction function is defined as:

$$d_e = G(d_0, \mathbf{I}, C, M, R) \qquad (1)$$

In Equation (1), I represent the outfit images or item-level visual features, C denotes clothing-insulation information, M denotes activity intensity and activity semantics, and R denotes the temperature reference value from simplified thermal mapping. These proxy variables primarily constrain weather and activity semantics so that insulation, activity intensity, and weather conditions in the generated description remain traceable. Style semantics are preserved and enriched from the original description and outfit visuals rather than inferred directly from the physical proxy variables.

3.1.2 Dataset and Experimental Split

We use the Polyvore Outfits Disjoint (PO-D) split as the primary experimental dataset. Earlier work established Polyvore-based fashion compatibility evaluation[14], whereas the disjoint split and its item-level separation were introduced by Vasileva et al.[26]. Compared with the non-disjoint version, the disjoint split prevents items from being shared across training, validation, and test sets, reducing the risk of item-level leakage and providing a stricter generalization setting. We retain the original PO-D split for comparability with prior outfit recommendation work. Without altering that split, we construct a context-aware textual description for every outfit. The construction process incorporates CLO, MET, temperature and thermal-comfort mapping, and occasion and style cues. After parsing and format validation, all 35,140 generated descriptions can be aligned with PO-D samples. The training, validation, and test sets contain 16,995, 3,000, and 15,145 outfits, respectively. The original Polyvore text source contains 68,306 records[26]. This number is reported only as the size of the available original-text source and is not the sample size of the main experiment.

To distinguish source-data size, usable generated samples, and recommendation-evaluation samples, Table 1 summarizes the dataset scope and major evaluation settings. The dataset-split and alignment rows describe the number of context-aware descriptions available to the main experiment, whereas the compatibility and recommendation rows describe the test samples actually used by the recommendation tasks. Compatibility evaluation includes 30,290 test pairs and 15,145 FITB questions. CIR evaluation requires a sufficiently large candidate pool for each target category. Consequently, 9,311 of the 15,145 test questions yield evaluable query-target pairs, while the remaining 5,834 are excluded from recall computation. Detailed task definitions and metrics are provided in Section 4.

Table 1: Dataset scope and evaluation settings

Category	**Item**	**Value / Description**
Dataset split	Train / validation / test / total outfits	16,995 / 3,000 / 15,145 / 35,140
Dataset alignment	Valid generated descriptions aligned with PO-D	35,140 / 35,140 (100.00%)
Original text records	Original Polyvore title/description records	68,306 records. Reported only as the number of available original text records, not as the main experimental sample size
Compatibility evaluation	Compatibility test pairs	30,290
Compatibility evaluation	FITB test questions	15,145
Recommendation evaluation	Evaluable query-target pairs per seed	9,311 (9,311 unique pairs)
Recommendation evaluation	Excluded from recall computation	5,834 test questions. Excluded by the task-specific eligibility filter

3.1.3 Data-Centric Motivation and Original-Text Limitations

Although PO-D provides outfit-level text, the original text primarily comes from webpage names, titles, or short thematic phrases and was not designed for context-aware outfit recommendation. Under our text-conditioned recommendation setting, the original outfit-level condition is defined as `url_name` + title. We therefore compare this original text condition with the generated context-aware description rather than using the title field alone as the baseline.

As shown in Table 2, among the 35,140 outfits aligned with the PO-D main experiment, only 34 original `url_name` + title queries are missing (0.10%), and the average length is 4.82 words. These texts remain mainly short phrases reflecting webpage names, style tags, or brief themes and provide limited information about thermal conditions, occasion requirements, activity intensity, or clothing insulation. By contrast, the generated context-aware descriptions have no missing values, average 10.08 words, and cover weather, occasion, and style.

The data-centric motivation arises not from missing original text but from its limited contextual coverage. The goal of rewriting is therefore not arbitrary text expansion. It is to transform short, commerce-oriented outfit text into traceable and auditable conditions that reflect everyday dressing needs. Because the generated descriptions are longer than the original text, subsequent experiments explicitly control for text length and audit target-clue leakage to reduce the risk of attributing performance gains merely to longer queries or exposed target-item clues.

Table 2: Comparison between original text fields and generated context-aware descriptions

Text field	Missing records	Word length (mean / median / max)	Mean query tokens	<=3 words
Original query	34 (0.10%)	4.82 / 5.00 / 25 words	4.90	21.88%
Context-aware description	0 (0.00%)	10.08 / 10.00 / 27 words	11.60	0.00%

3.1.4 Framework Overview

The overall method consists of three components: context-aware description construction, generation-quality diagnostics, and downstream recommendation evaluation. First, outfit images and original text from PO-D are combined with CLO and MET references to estimate outfit-level insulation and plausible activity intensity. Second, McIntyre's simplified thermal mapping converts CLO and MET into a temperature reference value used to constrain weather semantics. Finally, a LLM rewrites the original short text into a context-aware description containing weather, occasion, and style semantics based on the original description, CLO/MET proxies, and the temperature reference value.

Figure 1 summarizes the context-aware description-construction pipeline. The process is designed to build learnable contextual text conditions from traceable proxy variables rather than to expand text freely. Each generated description can be traced to the original outfit images and text, estimated CLO, activity/MET, temperature reference value, and prompt configuration, making the construction process easier to inspect and explain than unconstrained rewriting.

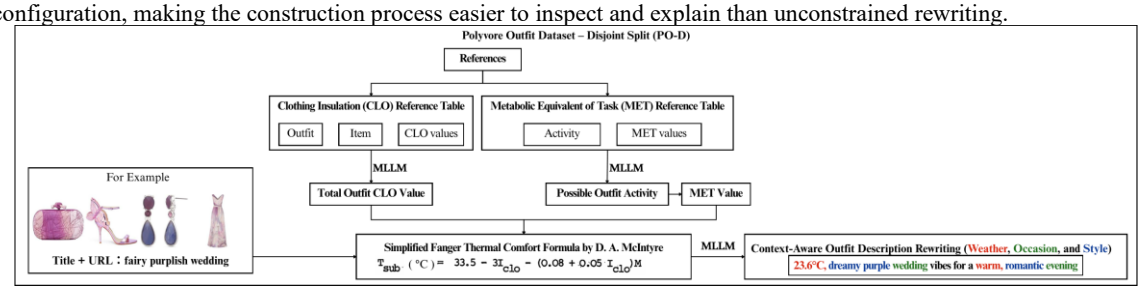


Figure 1: Context-aware outfit description construction pipeline using CLO/MET proxy variables and thermal mapping.

After description construction, checklist-based LLM-as-a-Judge diagnoses generation quality by checking image-text consistency, representation of contextual conditions, and semantic drift. These diagnostics are not used as a direct measure of recommendation performance. They evaluate whether the generated descriptions are usable and auditable. The constructed descriptions are then provided to the Hybrid Attention text-conditioned outfit recommendation model and compared with the original text under identical data splits, visual representations, model architecture, and training settings, isolating the source of the text condition. A simpler Two-Tower architecture also provides cross-architecture robustness validation and serves only as a complementary validation model rather than replacing the primary Hybrid Attention setting.

3.2 Context-Aware Description Construction

This subsection describes the construction pipeline, from CLO estimation and MET mapping to thermal-reference calculation, contextual rewriting, and reproducibility controls.

3.2.1 CLO Estimation and Clothing-Insulation Sources

Clothing insulation (CLO) is a standardized measure of the thermal resistance provided by clothing to heat flow from the body. The commonly used conversion is 1 clo = 0.155 $m^2 \cdot K/W$. We use CLO's comparable garment-insulation scale as an estimated outfit-level proxy for weather-semantic generation and thermal mapping, not as a measured human-comfort label [9, 11].

To establish a traceable CLO reference basis, we compile three sources of clothing-insulation data: ensemble-level insulation data from ASHRAE RP-1760, modern empirical studies of item- and ensemble-level insulation, and clothing-insulation lookup values associated with ASHRAE Standard 55[2, 23, 30]. These sources are organized into item-level and ensemble-level CLO reference tables that support LLM-based estimation of outfit-level CLO from outfit images and item information.

Because ensemble CLO is not necessarily a linear sum of item-level values, we retain item- and ensemble-level references. Layering, inter-garment air layers, coverage area, material thickness, garment length, and design details can all influence overall insulation. We therefore treat CLO as an estimated proxy that provides a traceable connection among outfit insulation, weather semantics, and text-condition construction. After estimation, output format and numeric range are checked, and the interquartile range (IQR) rule is used to flag outlying estimates for re-evaluation. Flagged records are re-inferred, not truncated or removed, preserving data-construction traceability.

3.2.2 MET Mapping and Activity-Constraint Construction

Metabolic Equivalent of Task (MET) represents metabolic intensity across activities and can serve as an individual parameter in thermal-comfort estimation. In general, 1 MET corresponds to the baseline metabolic level of seated rest. Following the conventional unit definition, 1 MET = 50 $kcal \cdot h^{-1} \cdot m^{-2}$ = 58.2 $W \cdot m^{-2}$[11]. In this study, activity-specific MET values are converted to metabolic rate in $W \cdot m^{-2}$ before Equation (2) is applied. MET characterizes the plausible activity intensity associated with an outfit and, together with CLO, serves as an intermediate scale for subsequent weather-semantic generation and thermal mapping.

To map activity semantics to MET values, we use the 2024 Adult Compendium of Physical Activities as the primary reference. The compendium associates textual activity descriptions with MET values and covers everyday activities such as walking, commuting, work, leisure, and household tasks, providing a standard reference for converting natural-language activity descriptions into metabolic intensity[15]. The original table contains 1,114 activities. To reduce prompt length and instability caused by redundant candidates, we sort by MET value and remove duplicate entries with identical MET values within major activity categories, yielding 457 representative activity candidates.

During generation, the model does not freely generate a MET value. Instead, it first selects the activity that best matches the outfit context from the curated candidate list and then maps that activity to its corresponding MET value. This design reduces numeric drift from free-form generation and preserves a traceable link among activity semantics, MET values, and the subsequent temperature reference value. MET is therefore not a measured activity level in this study, but a proxy used to constrain activity semantics and thermal mapping.

3.2.3 Simplified Thermal Mapping

After obtaining outfit-level CLO and MET, we use McIntyre's simplified relationship based on Fanger's thermal-comfort model to map clothing insulation and activity intensity to a temperature reference value[9, 21]:

$$T_{\mathrm{sub}} = 33.5 - 3I_{\mathrm{cl}} - (0.08 + 0.05I_{\mathrm{cl}})M \qquad (2)$$

In Equation (2), I_{cl} denotes clothing insulation in clo, M denotes metabolic rate in $W \cdot m^{-2}$ after the above MET conversion, and T_{sub} denotes the temperature reference value used for data construction. Under this simplified proxy setting, external mechanical work is assumed negligible. The mapping converts numeric CLO and MET annotations into a weather-semantic reference expressible in natural-language descriptions.

Importantly, we do not use this equation as a complete Predicted Mean Vote (PMV) calculation, nor do we interpret T_{sub} as the actual comfort temperature for every user. A complete thermal-comfort assessment generally also requires air temperature, mean radiant temperature, air velocity, humidity, individual differences, and clothing configuration, none of which are measured in this dataset. Accordingly, T_{sub} is used only as a data-construction variable that establishes a relative mapping among clothing insulation, activity intensity, and weather semantics.

Simplified thermal mapping produces more consistent weather and temperature semantics across CLO/MET combinations. Holding the other variable constant, greater clothing insulation or higher activity intensity maps to a lower temperature reference, whereas lower clothing insulation or lower activity intensity maps to a higher temperature reference. This relationship reflects the compensatory roles of clothing insulation and metabolic heat production in the simplified mapping. The objective is not to predict actual human thermal comfort, but to reduce reliance on unconstrained weather generation and preserve a traceable relationship among insulation, activity intensity, and environmental semantics in the context-aware descriptions.

3.2.4 Context-Aware Description Rewriting

The rewriting stage transforms short outfit-level text into query-like natural-language conditions that jointly express weather, occasion, and style semantics. We use Google Gemma 3 4B IT (gemma-3-4b-it)[1] for rewriting because it supports multimodal input and large-scale data construction [12]. Model inputs include the original description, outfit images, the outfit-level CLO estimate, the MET-based activity-intensity reference, and the temperature reference value. The MET-linked activity candidate is not the final occasion label; it constrains activity intensity so that generated weather and occasion semantics do not clearly conflict with outfit insulation or metabolic intensity. The semantic content of the final description remains grounded primarily in the original text and outfit images.

For example, the original description “fairy purplish wedding” can be rewritten as “23.6°C, dreamy purple wedding vibes for a warm, romantic evening.” Here, Weather is represented by “23.6°C” and “warm,” Occasion by “wedding” and “evening,” and Style by “dreamy,” “purple,” and “romantic.” This example illustrates that rewriting transforms a short source tag into a text condition containing environmental, occasion, and styling semantics rather than merely expanding it arbitrarily.

The generated output is retained as a single natural-language description rather than being separated into independent Weather, Occasion, and Style fields. This design reflects how users typically express mixed semantics in outfit searches, where temperature, activity or occasion, and overall style often appear in the same query. Weather/Occasion/Style (W/O/S)

[1] https://huggingface.co/google/gemma-3-4b-it

factors are used later to construct the fair subset, perform semantic-factor ablations and counterfactual sensitivity analysis, and interpret results. Model training still uses the integrated natural-language description as the text condition to match the input form of a text-conditioned outfit recommender.

3.2.5 Generation Constraints, Traceability, and Reproducibility

To reduce formatting instability and semantic drift during LLM generation, we fix the input sources, reference data, prompt version, and post-processing rules during data construction. CLO estimation uses fixed clothing-insulation reference tables and outfit information. MET mapping is restricted to a curated activity candidate list. Thermal mapping uses the fixed McIntyre simplification, and context-aware rewriting uses a fixed model and prompt template. These choices constrain free-form generation to a repeatable process rather than arbitrary model-generated contextual conditions.

Generated outputs retain their correspondence with the original outfit data and include intermediate fields for CLO, MET, the temperature reference value, and the final context-aware description. These fields enable later sample-level analysis of relationships among proxy values, textual descriptions, and recommendation results. The traceability claimed here derives primarily from fixed data sources, fixed rules, and sample-level field alignment rather than from a fully automated data-versioning system.

Prompt refinement is part of method development, not a formal ablation. Early prompt versions were used to correct unstable output formats, numeric parsing difficulties, overgeneralized activity semantics, and insufficient representation of contextual conditions. Only the final fixed prompt was used for formal data construction. Because different prompt drafts were not used to retrain and reevaluate recommendation models under otherwise identical conditions, we do not claim quantitative superiority among prompt versions. Instead, we report the final construction rules, main downstream recommendation evaluation, and complementary robustness analyses.

3.3 Generation-Quality Diagnostics

This subsection explains the checklist-based LLM-as-a-Judge procedure and manual audit used to diagnose generation quality without filtering the recommendation data.

3.3.1 Checklist-Based LLM-as-a-Judge Procedure

Context-aware description rewriting has no unique reference answer. The same outfit images and original description can support multiple reasonable formulations, so fluency or one holistic score cannot adequately capture differences in image-text consistency, representation of contextual conditions, and semantic reliability. We therefore adopt a checklist-based LLM-as-a-Judge procedure that decomposes description quality into itemized criteria for diagnosing image-text inconsistency, missing contextual conditions, semantic drift, and generated content unsupported by the images.

As shown in Figure 2 , the checklist covers three dimensions: content and context, visual attributes, and text quality. Items include weather or temperature, occasion or usage, overall style, color, visible design features, and hallucination control. We weight checklist items by visual grounding, contextual representation, and text quality, giving higher priority to image-text consistency and hallucination control.

ID	Assessment Criteria	Check
Dimension 1: Content & Context		
c1	Does the description include weather/temperature context?	☐
c2	Does it mention the overall style (e.g., evening glamour)?	☐
	.....	
Dimension 2: Visual Attributes		
c6	Does it accurately reflect the color palette?	☐
c7	Does it mention key design features (e.g., lace)?	☐
	.....	
Dimension 3: Quality Control		
c8	No Hallucinations (avoids non-existent elements)?	☐
c10	Is the text grammatically correct and fluent?	☐

ID	Wgt.	Assessment Criteria	Check
Tier 1: High Priority (Safety & Grounding)			
c8	4	No hallucinations (avoids non-existent elements)	☐
Tier 2: Content & Context			
c1	3	Includes temperature/weather context	☐
c4	3	Conveys intended occasion or context	☐
		.....	
Tier 3: Visual & Text Quality			
c6	2	Reflects accurate color palette	☐
c10	1	Grammatically correct and fluent	☐
		.....	

Figure 2: Checklist-based diagnostic criteria for generated context-aware outfit descriptions.

Algorithm 1 presents the three-stage scoring procedure used in this study. The procedure adapts PREF's staged rubric construction and Check-Eval's checklist decomposition to the multimodal outfit-description setting [10, 22]. Phase 1 constructs a general checklist. Phase 2 adapts the criteria and weights to the context-aware outfit-description task. Phase 3 then scores each outfit image-description pair item by item and returns a quality score, feedback, and a suggested revision. The item-level evaluation is designed to examine whether each generated description is grounded in the corresponding outfit image and appropriately reflects the required contextual information. Using two judging models also enables us to examine the consistency of the evaluation across different multimodal models rather than relying on a single evaluator. We use two multimodal judging models, Qwen3-VL-32B-Instruct[2] and Gemma-3-27B-it[3], to evaluate all 35,140 context-aware descriptions [3, 12]. The diagnostics were used to assess whether systematic quality problems required dataset-level intervention. Because no such systematic problems were identified, no additional sample filtering or rewriting was applied, and the original batch-generated descriptions were retained for downstream recommendation evaluation.

ALGORITHM 1: Three-phase LLM-as-a-Judge procedure for generated-description diagnostics

Require:

Image collage set $I = \{i_n\}_{n=1}^{N}$

Description set $D = \{d_n\}_{n=1}^{N}$

Paired dataset $\mathcal{X} = \{\mathcal{X}_n\}_{n=1}^{N}$ with $\mathcal{X}_n = (i_n, d_n)$

Coverage LLM M_{cov}

Preference LLM M_{pref}

Scoring LLM M_{score}

Ensure:

[2] https://huggingface.co/Qwen/Qwen3-VL-32B-Instruct

[3] https://huggingface.co/google/gemma-3-27b-it

Scores $S = \{s_n\}_{n=1}^{N}$

Feedback $F = \{f_n\}_{n=1}^{N}$

Suggested Revision $R = \{r_n\}_{n=1}^{N}$

Phase 1: Generate General Guidelines

C ← GENERATEGUIDELINE(M_cov, X)

$C = \{\boldsymbol{c}_1, \ldots, \boldsymbol{c}_m\}$, a universal checklist of yes/no items (criterion-guided, reference-free).

Phase 2: Personalize Guidelines

C* ← PERSONALIZE(M_pref, X, C)

$C*$ re-ranks and may augment C according to global task conditions; denote $C^* = \{c_1^*, \ldots, c_{m*}^*\}$ (m* may differ from m). All inputs thereafter are evaluated against the same C*.

Phase 3: Score Descriptions

for each $\mathcal{X}_n = (i_n, d_n)$ in $\mathcal{X}$ **do**

(s_n, f_n, r_n) ← SCORE(M_score, X_n, C*)

end for

return **S**, **F**, **R**

3.3.2 Manual Audit and Diagnostic Use

LLM-as-a-Judge scores are used for generation-quality diagnostics rather than as training signals or sample-selection criteria for the primary recommendation model. Judge-produced feedback and suggested revisions are retained to analyze failure types and potential corrections for low-scoring samples. To complement the automated diagnostics with human review, we conduct a stratified manual audit. Formal quality scores from the two judges are converted to within-model percentile ranks and used to define high-score, low-score, and disagreement strata. Low-score samples have a mean percentile no greater than 0.25 and a percentile difference no greater than 0.35. High-score samples have a mean percentile of at least 0.75 and a percentile difference no greater than 0.35. Disagreement samples have a percentile difference of at least 0.50 or an absolute raw-score difference of at least 0.25. We sample 10 cases from each stratum, yielding 30 manual-audit samples under a fixed random seed. The sample set also covers diverse temperatures, activity intensities, occasions, and styles. Judge scores are not used to delete or revise generated samples directly. Feedback and suggested revisions are retained only for diagnostics, and the formal downstream recommendation evaluation uses the original batch-generated context-aware descriptions so that an additional revision stage does not alter the main data condition.

The manual audit uses the weighted checklist defined for each judging model. Given the same outfit images and description, the researcher makes a binary decision per checklist item and calculates a human score with the automated evaluation weights. Judge A and Judge B contain 10 and 15 checklist items, respectively, producing 750 human item-level decisions across the 30 samples. To compare automated scores with human judgments, we report mean bias, mean absolute error (MAE), root mean square error (RMSE), Pearson correlation, and Spearman rank correlation. The manual audit is used only as complementary validation of LLM-as-a-Judge reliability. It is not treated as a large-scale gold standard and is not used to rescreen or modify recommendation-model training data.

3.4 Downstream Recommendation Evaluation

This subsection specifies the controlled downstream recommender and the W/O/S annotations used for semantic-factor analysis.

3.4.1 Primary Recommendation Model

The primary recommendation model is Wang and Zhong's Hybrid Attention text-conditioned outfit model[28]. The model uses FashionCLIP [5] to extract item image and text representations and integrates textual conditions with candidate-item features through the Hybrid Attention Layer to estimate candidate fit with the existing outfit and textual requirement.

As shown in Figure 3, generated context-aware descriptions are used as textual conditions while item images, item names, feature extraction, model architecture, and data splits are held constant. This design controls recommendation-model factors so that experimental differences primarily reflect the source of the textual condition. The comparison therefore focuses on original `url_name` + title queries versus generated context-aware descriptions within the same text-conditioned recommendation framework.

We do not modify the core Hybrid Attention Layer or claim a new recommendation-model design. The main experiments hold the recommender, multimodal representations, data splits, and training settings fixed so that the comparison isolates text-condition source and tests whether context-aware descriptions improve outfit compatibility and conditional item retrieval. A simpler FashionCLIP-based Two-Tower architecture is additionally used for cross-architecture robustness validation. It is a complementary validation model and does not replace the primary recommendation setting.

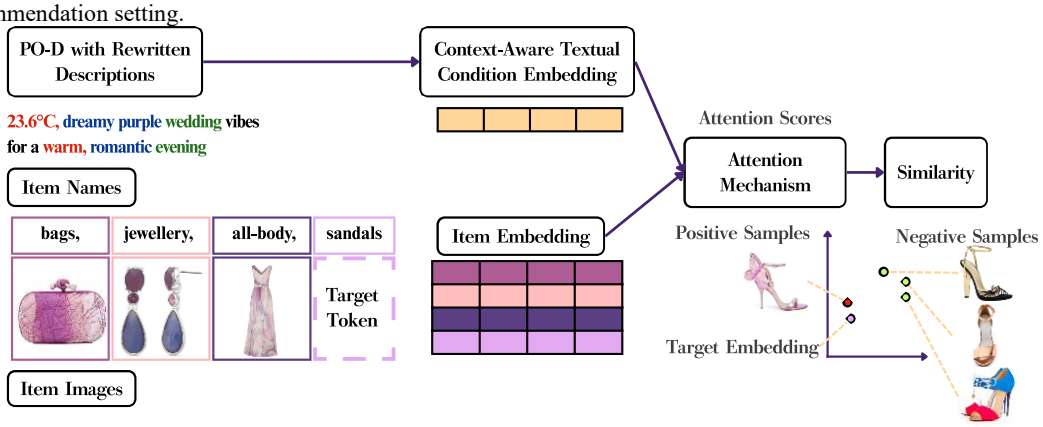


Figure 3: Controlled text-conditioned outfit recommendation framework with generated context-aware descriptions. Adapted from Wang and Zhong [28], with the textual condition revised to the generated context-aware description used in this study.

3.4.2 Contextual Semantic-Factor Annotation

Although training uses each context-aware description as one natural-language condition, later analyses identify whether it contains weather, occasion, and style semantics. We therefore annotate generated descriptions with Weather, Occasion, and Style (W/O/S) factors according to the operational definitions in Table 3. These annotations support fair-subset construction, semantic-factor ablation, contextual-subset comparisons, and counterfactual sensitivity analysis. They are not supplied to the recommendation model as additional structured features.

Table 3: Operational definitions of W/O/S semantic factors

Factor	Operational definition	Examples
Weather (W)	Describing external climate, temperature, season, or thermal-feeling context. Hard indicators include temperature values, weather phenomena, and seasonal terms. Soft indicators include weather-related feeling terms such as warm, cool, and chilly.	23.6°C, warm, chilly, summer

Factor	Operational definition	Examples
Occasion (O)	Text spans that independently express an activity, place, time context, or usage scenario. Only explicit context phrases are counted. If a phrase only implies a possible context through style but does not contain an identifiable occasion cue, it is not counted as Occasion.	date, travel, party, walk, evening
Style (S)	Text spans describing fashion style, aesthetic direction, garment type, material, pattern, or visible styling cues. These terms indicate the visual or stylistic expression of the outfit rather than a user's personal identity or long-term preference.	casual, formal, heels, floral prints

This annotation establishes an analyzable semantic layer; it does not decompose the description and re-enter its factors separately into the model. The recommendation model still receives the complete natural-language description, while W/O/S annotations are used to characterize semantic coverage and support the fair-subset, ablation, contextual-subset, and counterfactual analyses.

4 EXPERIMENTAL DESIGN

This section specifies the evaluation tasks, controlled comparison, training settings, semantic-factor ablations, and robustness analyses used to test the constructed descriptions.

4.1 Tasks and Evaluation Metrics

We evaluate the effect of context-aware textual conditions on multimodal outfit recommendation using compatibility and conditional recommendation tasks. Outfit compatibility includes Compatibility Prediction (CP) and Fill-in-the-Blank (FITB). CP evaluates the model's ability to distinguish compatible from incompatible outfits and is measured primarily by area under the receiver operating characteristic curve (AUC). FITB evaluates whether the model can select the correct missing item from a candidate set and is reported as accuracy.

Conditional recommendation is evaluated with Conditional Item Retrieval (CIR). Given a text condition and a partial outfit, CIR measures whether the correct target item is ranked near the top of the candidate list. Let N be the number of queries eligible for CIR evaluation and r_i the rank of the correct target item for query i. Recall@K is defined in Equation 3:

$$\text{Recall@}K = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\,(r_i \le K) \qquad (3)$$

where $I(\cdot)$ is the indicator function and equals 1 when the correct target item appears within the top K candidates and 0 otherwise. We report Recall@1, Recall@3, Recall@5, Recall@10, Recall@30, and Recall@50 to characterize retrieval performance at different ranking depths.

The main experiments are repeated with five fixed random seeds, and we report the mean, standard deviation, and 95% confidence interval (CI). Because original text and context-aware descriptions are compared under the same random seeds and experimental settings, the observations are paired. We use paired t-tests to assess differences between the two text conditions and paired-sample Cohen's d_z to quantify standardized effect sizes. The eight primary recommendation metrics—AUC, FITB Accuracy, Recall@1, Recall@3, Recall@5, Recall@10, Recall@30, and Recall@50—are treated as a single family, and the p-values are jointly adjusted using the Benjamini–Hochberg procedure.

Let D_i denote the metric difference between the context-aware description and the original text for random seed i, and let $\bar{D}$ and s_D denote the mean and standard deviation of these differences. Cohen's d_z is defined in Equation 4:

$$d_z = \frac{\bar{D}}{s_D} \qquad (4)$$

A positive difference indicates that the context-aware description outperforms the original text on the corresponding metric. Results are interpreted jointly using the mean difference, 95% CI, Cohen's d_z, and Benjamini–Hochberg-adjusted paired t-test p-values.

4.2 Comparison Conditions and Control Analyses

The primary comparison contrasts original outfit-level text (url_name + title) with generated context-aware descriptions. Both conditions use the same data splits, item images, item names, feature extraction, recommender architecture, and training settings. Thus, the main experiment isolates the text-condition source's influence on recommendation performance.

To examine whether performance differences could be explained by other factors, we include three control analyses. First, a text-length check analyzes the relationship between query-length differences and retrieval improvement to determine whether gains are attributable only to longer generated descriptions. Second, a Target-Clue Leakage Audit examines whether generated descriptions systematically expose target categories or target-item cues. Third, we inspect the distributions and plausibility of CLO, MET, and temperature reference values to verify that the proxy variables remain within interpretable ranges during data construction. Accordingly, this comparison evaluates the overall effect of the constructed context-aware text condition and does not independently isolate proxy guidance from generic LLM-based rewriting.

4.3 Experimental Settings

Data preprocessing and early rewriting validation used a Google Colab A100 GPU. Subsequent LLM validation and recommender training used an NVIDIA RTX PRO 6000 Blackwell Workstation Edition GPU with 96 GB VRAM. All primary comparisons use the same data splits and FashionCLIP representations to minimize confounding from preprocessing or feature extraction.

The CP task trains the fixed Hybrid Attention recommender using FashionCLIP item representations. The model is trained for 100 epochs with a batch size of 50 and a learning rate of 5×10^{-5}. Hard negative sampling starts after epoch 40 to better distinguish difficult negative examples.

For CIR, CP-stage model weights initialize further conditional complementary-item retrieval training. This setup follows the text-conditioned outfit recommendation pipeline in which the model first learns outfit compatibility and is then adapted for target-item ranking. We compare Recall@K under original outfit-level text and context-aware descriptions to quantify the contribution of contextual text conditions to conditional outfit recommendation.

4.4 Contextual Semantic-Factor Ablation

In addition to the main experiment, we use the W/O/S annotations from Section 3.4.2 to design contextual semantic-factor ablations that assess the effects of weather, occasion, and style semantics on recommendation performance. The main experiment compares Original text with Context-aware description using all valid samples. The ablation uses a fair subset in which Weather, Occasion, and Style are all present, preventing differences in semantic coverage from confounding interpretation. Four text conditions are constructed: Full/Context-aware description, No-Weather (No-W), No-Occasion (No-O), and No-Style (No-S). Full retains the complete context-aware description, whereas No-W, No-O, and No-S remove weather-, occasion-, and style-related spans, respectively. All ablations use identical data splits, feature extraction, and recommender settings so that differences are concentrated in the contextual semantics retained in the text.

4.5 Robustness Validation Design

Robustness is assessed through cross-architecture replication, counterfactual text controls, and repeated-seed significance testing.

4.5.1 Cross-Architecture Validation

To assess whether the benefits observed in the main experiment depend strongly on Hybrid Attention, we additionally evaluate a simpler FashionCLIP-based Two-Tower model. The validation uses the same Polyvore Outfits Disjoint split as the main experiment and independently trains the model with original outfit text and context-aware descriptions, keeping text source as the principal difference between conditions. The Two-Tower model serves only as a complementary robustness architecture and does not replace the Hybrid Attention model used in the main experiment.

FashionCLIP first produces separate 512-dimensional image and text representations for each item. These are concatenated into a 1,024-dimensional item representation and transformed by a shared projection layer into a 256-dimensional feature. For CP, pooled item representations and the outfit-level text representation are jointly used for compatibility prediction. For conditional retrieval, the partial outfit, text condition, and target category form the query representation, which is compared with candidate-item representations to produce rankings. The retrieval stage reuses CP-trained item and text encoders, preserving the sequence from compatibility learning to candidate ranking.

Original text and context-aware descriptions are each trained under five fixed random seeds. The CP and retrieval models are trained for 100 epochs with a batch size of 256, learning rate of 1×10^{-3}, AdamW optimizer, weight decay of 1×10^{-4}, and dropout of 0.15. The model checkpoint with the best validation FITB accuracy is selected.

4.5.2 Counterfactual Context Sensitivity Analysis

To further assess whether recommendation rankings respond to changes in contextual semantics, we design a single-factor counterfactual context sensitivity analysis. Rather than regenerating the entire description or changing multiple contextual conditions simultaneously, we replace only one semantic factor—Weather, Occasion, or Style—while holding all other inputs and retrieval settings fixed, and observe the sensitivity of candidate rankings to that controlled semantic change.

Let the original context-aware text condition for sample i be denoted in Equation 5:

$$q_i = (W_i, O_i, S_i) \quad (5)$$

where $W_i, O_i,$ and S_i denote the Weather, Occasion, and Style semantics in the text condition. We construct the single-factor counterfactual conditions defined in Equations 6–8:

$$q_i^{(W)} = (W_i', O_i, S_i) (6)$$

$$q_i^{(O)} = (W_i, O_i', S_i) (7)$$

$$q_i^{(S)} = (W_i, O_i, S_i') (8)$$

where W_i', O_i', and S_i' denote the counterfactual semantics after factor replacement. Each counterfactual operation modifies only the target factor while preserving the other two contextual components, reducing interference from simultaneous multidimensional changes.

Counterfactual cases are sampled from the 9,311 query-target pairs in the formal CIR test for which a complete candidate pool can be constructed. Each case retains the same partial outfit, target item, target category, and candidate pool.

As in formal CIR evaluation, the pool contains 3,000 candidate items from the same fine-grained category as the target. Sampling uses random seed 42 and avoids selecting the same outfit more than once. We construct eight cases each for Weather, Occasion, and Style. Each factor has two opposing perturbation directions, with four cases sampled per direction, yielding 24 controlled counterfactual pairs. We also balance target types as closely as possible between clothing-led and accessory-led items, resulting in 12 cases of each type.

Weather perturbations use the evaluable samples' median temperature reference of 22.9°C as the dividing point. Samples at least 2°C above or below the median represent warmer or colder conditions, respectively, producing warm-to-cold and cold-to-warm perturbations. Occasion perturbations use the formal/casual semantics in the existing descriptions to construct formal-to-casual and casual-to-formal replacements. Style perturbations are defined according to semantic richness: high richness indicates at least two identifiable style terms, whereas low richness indicates no more than one, producing high-to-low and low-to-high replacements. These controlled replacements are used to examine whether candidate rankings respond to explicit changes in contextual semantics rather than to determine whether the modified condition is inherently better or more appropriate. Table 4 summarizes the operational settings and sample allocation.

Table 4: Counterfactual perturbation settings for contextual semantic factors

Factor	Perturbation direction	Operational setting	n
Weather	warm → cold	Replace the warmer condition with a colder-weather description emphasizing insulation and layered dressing.	4
	cold → warm	Replace the colder condition with a warmer-weather description emphasizing lightweight and breathable dressing.	4
Occasion	formal → casual	Replace formal-occasion semantics with everyday casual-activity semantics.	4
	casual → formal	Replace casual-occasion semantics with formal work-context semantics.	4
Style	high → low style richness	Replace the richer style description with simpler styling semantics of lower style density.	4
	low → high style richness	Replace the simpler style description with richer, more layered styling semantics.	4
Total			24

During retrieval, only the modified text description is re-encoded. The partial outfit, target item, target category, candidate pool, and recommender remain fixed. We use the fixed weights of the existing context-aware CIR model and do not retrain it on counterfactual text, so that ranking differences between the original context-aware and counterfactual conditions are concentrated as much as possible on the changed textual context.

To quantify ranking responses after counterfactual replacement, we first calculate the target-item rank change. Let ri be the target rank under the original context-aware condition and ri' the rank under the counterfactual condition. The rank difference is defined in Equation 9:

$$\Delta Rank_i = r_i' - r_i \quad (9)$$

A positive ΔRanki indicates that the target moves down after counterfactual replacement; a negative value indicates that it moves up. Because the counterfactual text has no new human-annotated recommendation ground truth, rank difference is used only to describe the direction and magnitude of ranking response. Its sign does not directly indicate improved or degraded recommendation quality.

In addition to target-item rank, we calculate the Top-1 changed rate to determine whether replacing a contextual factor changes the highest-ranked candidate. Let $T1(q_i)$ and $T1(q_i')$ denote the top-ranked candidate under the original context-aware and counterfactual conditions, respectively. The Top-1 change for a single case is defined in Equation 10:

$$\text{Change@1}_i = \mathbb{I}[T_1(q_i) \neq T_1(q_i')] \quad (10)$$

where I(·) is the indicator function and equals 1 when the top candidate differs before and after counterfactual replacement and 0 otherwise. The average across all cases is the Top-1 changed rate, representing the proportion of counterfactual perturbations that change the first recommendation.

To measure broader changes among high-ranked candidates, we also compute the Jaccard overlap between the Top-5 candidate sets under the original and counterfactual conditions. Let $T5(q_i)$ and $T5(q_i')$ denote the two Top-5 candidate sets. The overlap is defined in Equation 11:

$$J_i^{(5)} = \frac{|T_5(q_i) \cap T_5(q_i')|}{|T_5(q_i) \cup T_5(q_i')|} (11)$$

The resulting J_i^5 lies between 0 and 1. Values closer to 1 indicate more similar Top-5 candidate sets after replacement and thus greater stability to contextual change. Lower values indicate greater reorganization of the high-ranked candidate set caused by the single-factor perturbation.

This analysis measures candidate-ranking sensitivity to a controlled change in one contextual semantic factor, not whether the original or counterfactual condition yields better recommendation quality. Because no corresponding human recommendation ground truth exists after the counterfactual change, we do not assume that the target rank should improve or decline, and we do not interpret rank differences as causal effects of contextual factors on recommendation performance. The analysis complements the W/O/S ablation by testing whether candidate rankings respond observably when other retrieval conditions are held fixed and one textual contextual factor is changed.

5 EXPERIMENTAL RESULTS

This section consolidates validity checks before reporting the primary Hybrid Attention results, semantic-factor validation, cross-architecture and counterfactual robustness, and the main applicability boundaries. Detailed defensive analyses are summarized only to the extent needed to support interpretation of the recommendation results.

5.1 Validity Checks for Generated Context-Aware Descriptions

Before downstream recommendation evaluation, we performed proxy-value sanity checks, text-length confounding analysis, a target-clue leakage audit, LLM-judge consistency diagnostics, prompt-robustness checks, and a stratified 30-sample manual audit. Table 5 consolidates the principal evidence. These checks assess whether the generated descriptions are usable as text conditions; LLM-based scores remain diagnostic signals rather than human ground truth.

Table 5: Summary of generated-data validity checks

Check	Key evidence	Interpretation
Proxy values	CLO: median 0.93; Q1–Q3 0.63–1.21; P5–P95 0.25–2.07; max 5.20; 20 records > 4. MET: median 1.00; Q1–Q3 1.00–1.30; P5–P95 1.00–1.50; 25,786 records (73.4%) at 1.0; 8 records > 10. Temperature: median 22.9°C; Q1–Q3 19.9–24.5°C; P5–P95 13.5–27.1°C; 257 records (0.73%) < 0°C; 51 (0.15%) < −20°C.	The central distributions are interpretable; extreme values are retained as proxy-boundary diagnostics.
Text length	N = 9,311; Pearson r = 0.027647 for ΔHit@10 and 0.043384 for rank improvement. Within five tokens (n = 3,014), ΔHit@10 = +0.0106.	Length has negligible association with the retrieval gain, which remains positive in matched-length samples.

Check	Key evidence	Interpretation
Target clues	Fine-category match 4.50%; any-category match 4.88%; target bigram overlap 0.87%; generated token overlap 11.06% versus 12.40% for original text.	Generated descriptions show limited explicit clues and no clear increase over the original condition.
Judge agreement	At the bottom 5%, 668 samples overlap versus 87.9 expected by chance (lift 7.604; Jaccard 0.2347; F1 0.3802).	The judges share non-random diagnostic signals but are not treated as human ground truth.
Prompt robustness	On 150 samples, P0-R2 mean absolute differences are 0.0048 (Judge A) and 0.0461 (Judge B); 97.3% and 89.3% are within 0.05 and 0.10, respectively.	Repeated scoring is relatively stable, while prompt rewording remains a source of variation.
Manual audit	Thirty stratified samples. Human/model means: 0.7242/0.6667 (Judge A) and 0.6818/0.6566 (Judge B); MAE 0.2273 and 0.2212.	Scores are useful for diagnosis, with residual disagreement requiring manual interpretation.

Overall, the checks support cautious use of the generated descriptions while retaining proxy boundary values and judge disagreement as limitations rather than filtering them post hoc.

5.2 Main Recommendation Results

All primary experiments use the same data split, FashionCLIP representations, Hybrid Attention recommender, and training settings; only the text source differs. Table 6 reports all eight primary metrics. The p-values shown in the table are paired t-test p-values jointly adjusted across the eight metrics using the Benjamini–Hochberg procedure.

Context-aware descriptions improve AUC by 0.0162 and FITB Accuracy by 0.0103. CIR also improves at every reported depth, including gains of 0.0141 at Recall@10 and 0.0215 at Recall@50. All 95% CIs exclude zero, and all improvements remain significant after Benjamini–Hochberg adjustment.

Table 6: Main results on compatibility prediction and conditional item retrieval

Task	Metric	Original text	Context-aware	Difference	95% CI	p	Cohen's d_z
CP	AUC	0.9292 ± 0.0013	0.9454 ± 0.0013	+0.0162	[0.0157, 0.0167]	< .001	38.934
CP	FITB Accuracy	0.6374 ± 0.0037	0.6478 ± 0.0021	+0.0103	[0.0054, 0.0153]	< .01	2.578
CIR	Recall@1	0.0139 ± 0.0004	0.0151 ± 0.0011	+0.0012	[0.00005, 0.00236]	< .05	1.292
CIR	Recall@3	0.0333 ± 0.0010	0.0369 ± 0.0020	+0.0036	[0.00003, 0.00714]	< .05	1.253
CIR	Recall@5	0.0476 ± 0.0010	0.0557 ± 0.0023	+0.0082	[0.00443, 0.01189]	< .01	2.719
CIR	Recall@10	0.0780 ± 0.0010	0.0921 ± 0.0035	+0.0141	[0.01006, 0.01816]	< .01	4.326
CIR	Recall@30	0.1677 ± 0.0045	0.1874 ± 0.0038	+0.0197	[0.01527, 0.02412]	< .001	5.526
CIR	Recall@50	0.2303 ± 0.0035	0.2518 ± 0.0038	+0.0215	[0.01606, 0.02686]	< .01	4.934

5.3 Semantic-Factor Validation and Ablation

We next use the fair subset to examine the contribution of each contextual factor while reducing the influence of coverage imbalance. Table 7 compares the Full and Original conditions with the No-Style ablation. Removing Style produces the largest and most consistent performance degradation, whereas removing Weather or Occasion results in smaller changes across the same metrics. These findings suggest that Style contributes most strongly to the observed recommendation gains under the current experimental setting, while Weather and Occasion provide smaller complementary benefits.

Table 7: Fair-subset validation and primary style-ablation effects

Metric	Original	Full	No-S	Full–Original	Full–No-S
AUC	0.9112 ± 0.0015	0.9265 ± 0.0015	0.9110 ± 0.0041	+0.0153	+0.0154

Metric	Original	Full	No-S	Full–Original	Full–No-S
FITB	0.6109 ± 0.0035	0.6228 ± 0.0034	0.6064 ± 0.0051	+0.0119	+0.0164
Recall@10	0.0686 ± 0.0043	0.0781 ± 0.0023	0.0671 ± 0.0035	+0.0096	+0.0110
Recall@30	0.1517 ± 0.0077	0.1720 ± 0.0034	0.1524 ± 0.0042	+0.0203	+0.0196
Recall@50	0.2111 ± 0.0046	0.2351 ± 0.0045	0.2110 ± 0.0014	+0.0241	+0.0241

Full remains superior to Original on all five fair-subset metrics. Removing Style reduces AUC, FITB, Recall@10, Recall@30, and Recall@50 by 0.0154, 0.0164, 0.0110, 0.0196, and 0.0241, respectively, indicating that style semantics contribute most consistently while weather and occasion provide smaller complementary constraints.

5.4 Cross-Architecture and Counterfactual Robustness

Table 8 provides supplementary cross-architecture validation with a simpler FashionCLIP-based Two-Tower model. CP AUC increases by 0.0378, while the FITB difference is −0.0077 and its confidence interval includes zero. Retrieval gains remain positive: Recall@10, Recall@30, and Recall@50 increase by 0.0102, 0.0153, and 0.0170, and mean rank decreases by 28.8911.

Table 8: Cross-architecture validation with a simplified FashionCLIP-based Two-Tower model

Metric	Original text	Context-aware description	Mean difference	95% CI
CP Test AUC	0.8796 ± 0.0029	0.9174 ± 0.0033	+0.0378	[0.0337, 0.0420]
CP FITB Accuracy	0.5460 ± 0.0051	0.5382 ± 0.0050	−0.0077	[−0.0180, 0.0025]
Recall@10	0.0378 ± 0.0008	0.0479 ± 0.0017	+0.0102	[0.0087, 0.0117]
Recall@30	0.0950 ± 0.0017	0.1103 ± 0.0030	+0.0153	[0.0127, 0.0179]
Recall@50	0.1419 ± 0.0026	0.1589 ± 0.0046	+0.0170	[0.0139, 0.0200]
Mean rank ↓	601.3589 ± 7.0359	572.4678 ± 10.5809	−28.8911	[−44.8960, −12.8863]
Median rank ↓	361.2000 ± 6.2209	329.8000 ± 9.7826	−31.4000	[−44.0808, −18.7192]

This validation does not position Two-Tower as a competing primary model; it tests whether the retrieval benefit is confined to Hybrid Attention. The largely consistent CIR direction suggests that the data-side semantic construction contributes beyond one recommendation architecture, although the mixed FITB result precludes a uniform cross-task claim.

The controlled counterfactual analysis holds the partial outfit, target item, category, candidate pool, and model fixed while replacing one semantic factor in 24 cases. Mean target rank changes from 586.0 to 603.0 (mean ΔRank +17.0), with a mean Top-5 Jaccard overlap of 0.351. Table 9 shows that Style replacement yields the largest ranking response, whereas Weather and Occasion effects vary in direction and magnitude.

Table 9: Counterfactual retrieval response overall and by contextual factor

Scope / factor	n	Mean ΔRank	Top-1 changed	Mean Top-5 Jaccard
Overall	24	+17.0	—	0.351
Weather	8	−146.4	0.12	0.508
Occasion	8	−11.9	0.38	0.416
Style	8	+209.1	1.00	0.130

Because no new human relevance labels exist after replacement, the sign of ΔRank indicates only movement of the existing target, not improvement or degradation under a new ground truth. The analysis therefore supports context sensitivity, not causal understanding.

5.5 Boundary Observations and Qualitative Cases

Performance varies by target category. As shown in Figure 4, dress, bag, and shoe categories exhibit the clearest gains, whereas sunglasses show a slight decline. The method therefore benefits items associated with outfit-level contextual semantics more consistently than accessories requiring fine-grained visual discrimination.

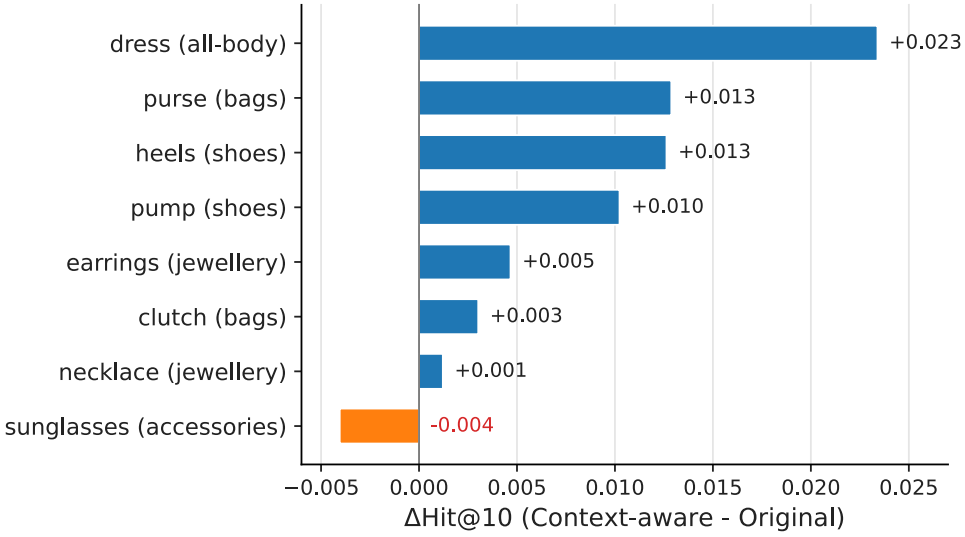


Figure 4: Hit@10 differences by item category.

Each label is shown as subcategory (main category), following the original Polyvore category taxonomy. Bars indicate absolute ΔHit@10 between context-aware and original text conditions.

Across 3,432 eligible query-target pairs (17,160 seed-level observations), context-aware descriptions also achieve higher Hit@10 in every reported weather, occasion, style-richness, and target-type subset; the weather split uses the median temperature reference of 22.9°C. The gain is therefore not confined to a single contextual condition, although its magnitude remains category-dependent.

Representative cases illustrate both benefits and limits. Mean target rank improves from 23.4 to 1.8 for a purse case and from 122.6 to 3.0 for a dress case; removing Style raises the dress-case mean rank to 60.2. In the color analysis, 39 of 89 row-level problems improve (43.82%), compared with 5 of 15 case-level problems (33.33%), so color alignment is not stable. A sunglasses failure case moves from mean rank 22.4 under Original text to 505.0 under the context-aware description, showing that broader context can dilute direct color or item cues. Additional retrieval examples are retained in Appendix A.1.

Overall, context-aware rewriting is most useful when retrieval depends on outfit-level semantics and less reliable when color, shape, or local design details dominate. These boundary observations qualify the aggregate gains without changing the primary conclusion.

6 DISCUSSION

This discussion interprets the results in four steps. Section 6.1 integrates the evidence for RQ1–RQ4 as research implications; Section 6.2 identifies where the benefits are most and least applicable; Section 6.3 consolidates the study's

interpretive scope, limitations, and residual evaluation risks; and Section 6.4 records the ethical, rights, and transparency considerations.

6.1 Synthesis of Key Findings

The findings support four implications corresponding to the research questions.

1. For RQ1, text-condition construction can improve a fixed recommender. With FashionCLIP representations, the Hybrid Attention architecture, data splits, and training settings held fixed, context-aware descriptions improve CP, FITB, and CIR. The Two-Tower validation shows positive changes in AUC and retrieval metrics but not a consistent FITB gain, indicating that the effect extends beyond one architecture while remaining task- and model-dependent. The matched-length analysis further suggests that the gains are unlikely to be explained by longer queries alone.
2. For RQ2, the contextual factors contribute unequally to recommendation performance. Removing Style causes the largest degradation in CP and CIR, while Style replacement produces the highest Top-1 changed rate and the lowest Top-5 Jaccard overlap. Weather and Occasion show smaller effects but provide complementary environmental and usage constraints. These results identify Style as the strongest contextual factor under the current setting, although the counterfactual evidence demonstrates ranking sensitivity rather than causal understanding.
3. For RQ3, CLO, MET, and temperature-reference values provide a traceable basis for contextual construction. Their sources, mappings, and distributions establish inspectable relationships among clothing insulation, activity intensity, and weather semantics. The central value ranges are interpretable for data construction, but these variables remain contextual proxies rather than individualized physiological measurements or thermal-comfort labels.
4. For RQ4, layered diagnostics support the controlled use of the generated descriptions in downstream recommendation evaluation. Dual-judge agreement, prompt-robustness checks, target-clue and text-length audits, and the stratified manual audit did not reveal systematic generation-quality problems requiring dataset-level intervention. Accordingly, no additional sample filtering or rewriting was applied after the diagnostic stage, and the original batch-generated descriptions were retained for downstream evaluation. Individual disagreements and failure cases remain documented as limitations, while LLM-based scores are treated as diagnostic signals rather than human ground truth.

6.2 Benefits and Applicability Boundaries

The benefit of context-aware descriptions depends on whether the retrieval target is governed mainly by outfit-level semantics or by fine-grained visual attributes.

1. The clearest benefits occur for dress, shoe, and bag targets. These categories depend more directly on overall style, formality, occasion, and season, so contextual descriptions provide useful discriminative cues.
2. Benefits are less stable for jewellery, accessories, and sunglasses, where color, shape, texture, and other local details dominate. The manual audit likewise locates the clearest disagreements in fine-grained image–text alignment rather than fluency.
3. Contextual semantics can dilute explicit source-text cues. In the failure case, broader sunset and beach semantics shifted retrieval toward candidates that matched the overall atmosphere but not the target's specific color and shape, while the color-shift analysis showed only limited improvement. The method is therefore most applicable when outfit-level context is central to relevance; targets whose relevance depends on fine-grained visual attributes require stronger attribute preservation and control.

6.3 Scope, Study Limitations, and Residual Evaluation Risks

The findings concern offline data construction and recommendation evaluation rather than direct user preference or wearing comfort. The main limitations are as follows.

1. Human and thermal validity. CP, FITB, and CIR do not measure satisfaction, willingness to wear, aesthetic preference, or actual comfort. CLO, MET, and temperature are traceable construction proxies rather than individualized physiological measurements or comfort labels; extreme temperature-reference values also expose limitations of the simplified thermal mapping.
2. Diagnostic validity. Two multimodal judges, prompt-robustness checks, and a stratified 30-sample manual audit support scalable generation-quality diagnostics, but individual disagreements remain. The scores can be influenced by prompt design, model priors, and visual-language capability and therefore cannot replace human annotation or real-user evaluation.
3. Model and dataset generalizability. The primary Hybrid Attention model and one complementary Two-Tower model are evaluated only on Polyvore Outfits Disjoint. Category distribution, source-text quality, temporal fashion bias, and the offline candidate pool limit generalization to other architectures, datasets, wardrobes, and interactive systems.
4. Residual experimental risks. Matched-length and target-clue analyses reduce but do not eliminate lexical confounding. Without a generic LLM-rewrite baseline or a masked-query CIR rerun, proxy-guided construction cannot be fully separated from general text enrichment or residual answer cues. The 24-case counterfactual analysis has no new human relevance labels and therefore supports ranking sensitivity, not improved quality or causal understanding.

6.4 Ethical, Rights, and Transparency Considerations

This study uses Polyvore Outfits Disjoint solely as a public benchmark for offline data construction and recommendation-model evaluation. It involves no human participants, surveys, user tracking, preference collection, or measurements of wearing comfort; accordingly, it makes no claims about real user preferences or thermal responses.

Public availability does not imply unrestricted redistribution rights for the original images. The paper therefore relies mainly on workflow diagrams, statistical and ranking tables, and only necessary qualitative examples.

AI use is reported for methodological transparency. Gemma-3-4B-IT generates context-aware conditions, while Qwen3-VL-32B-Instruct and Gemma-3-27B-IT perform checklist-based LLM-as-a-Judge evaluation. The diagnostic analyses did not identify systematic generation-quality problems requiring dataset-level intervention. Accordingly, no additional sample filtering or rewriting was applied after the diagnostic stage, and the original fixed batch-generated descriptions were retained for downstream recommendation evaluation. Individual disagreements and failure cases remain documented as limitations, while judge scores are treated as diagnostic signals rather than human ground truth.

7 CONCLUSION

This study presents a traceable data-side framework that converts short outfit text and images into weather-, occasion-, and style-aware conditions using CLO/MET proxies, simplified thermal mapping, fixed-prompt generation, and generation-quality diagnostics. With the primary Hybrid Attention setting controlled, context-aware descriptions improve CP, FITB, and CIR; the Two-Tower validation retains AUC and retrieval gains, and the ablation and counterfactual analyses identify Style as the strongest contextual factor.

The findings show that better contextual text can improve a fixed multimodal recommender without changing its backbone, but the benefit is strongest for targets governed by outfit-level semantics and weaker for targets governed by fine-grained visual attributes. The contribution is therefore a traceable and auditable semantic-construction pipeline with multi-level evaluation, not a claim of personalized comfort, universal model transfer, or causal understanding. Future work should add user studies, cross-dataset and multi-architecture validation, a generic rewrite baseline, masked-query CIR evaluation, and finer visual-attribute control.

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A APPENDICES

A.1 Qualitative Retrieval Examples

Figure A1, Figure A2, and Figure A3 provide additional qualitative examples for the CIR task. Each example compares the top retrieved candidates under Original text, No-Weather, No-Occasion, No-Style, and Context-aware description conditions. The green box marks the ground-truth target item when it appears in the displayed ranking. These examples are used only as qualitative references for the ranking behavior and applicability boundaries discussed in Section 5.5.

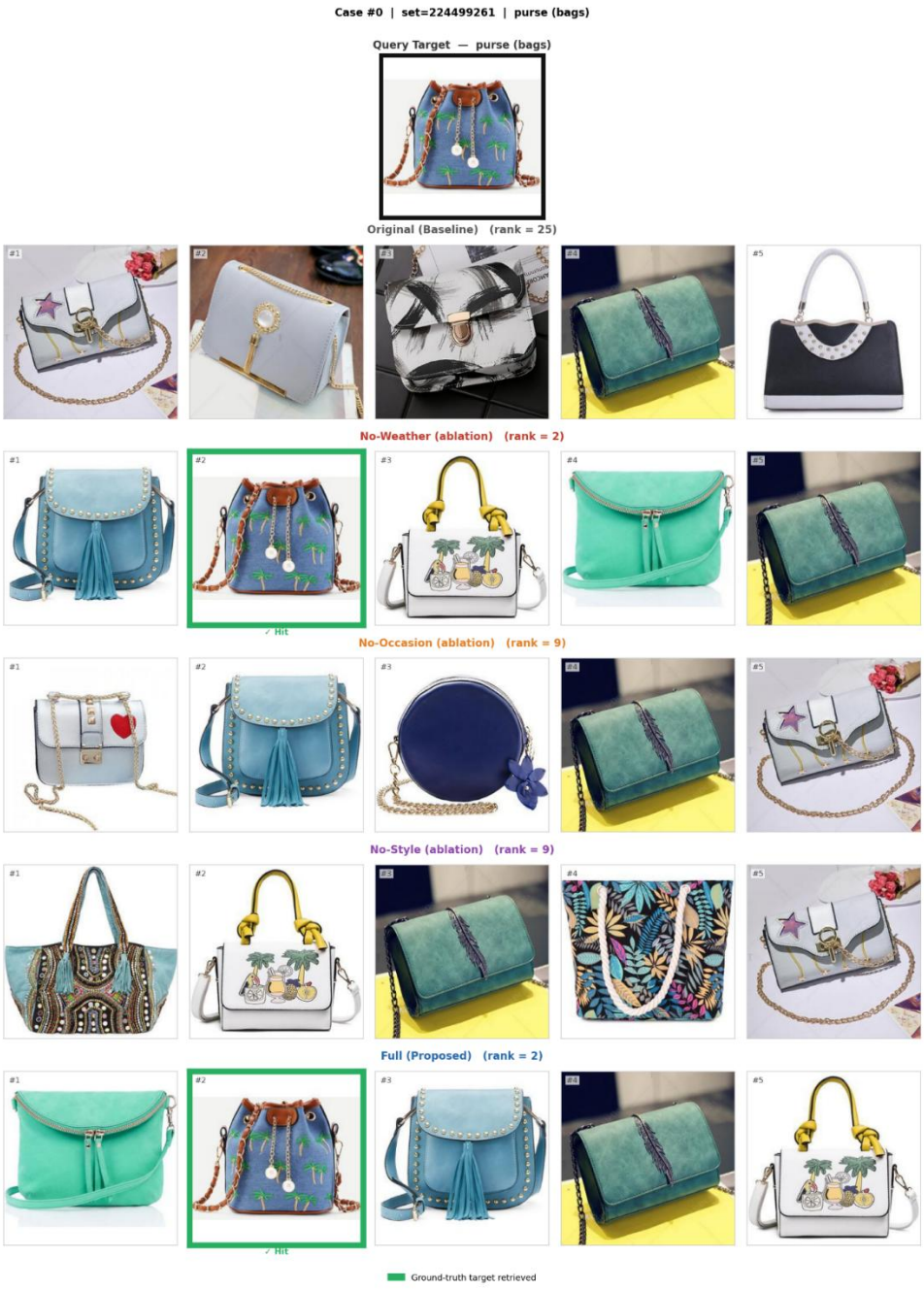


Figure A1: Qualitative retrieval example for a purse target item. The context-aware description retrieves the ground-truth purse at rank 2, whereas the original text condition ranks the target item at rank 25. The ablation results show that removing occasion or style information weakens the ranking compared with the full context-aware description.

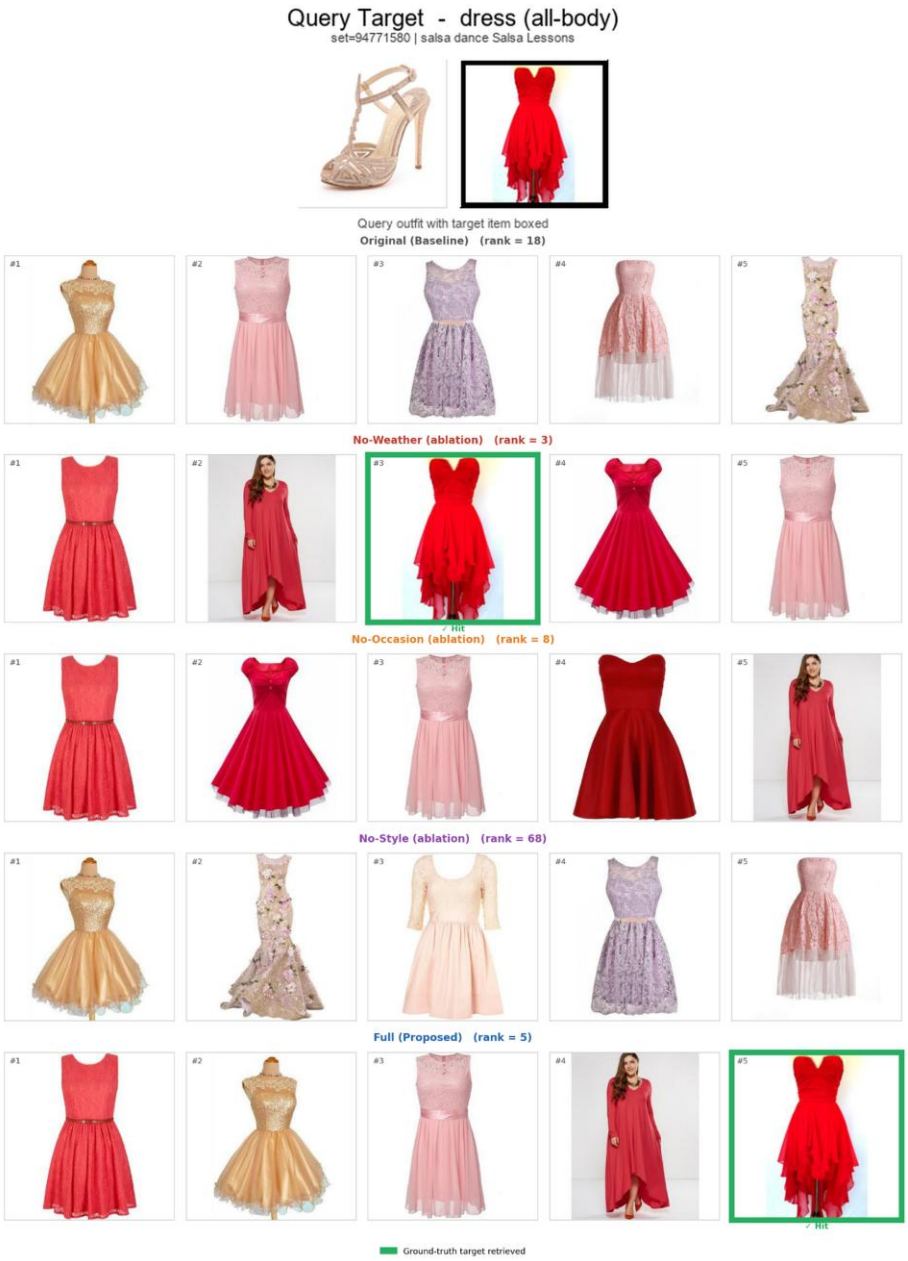


Figure A2: Qualitative retrieval example for a dress target item. The context-aware description improves the target dress ranking from the original condition and retrieves visually related red dress candidates in the top results. The No-Style condition shows a larger ranking degradation, consistent with the role of style semantics in dress-oriented retrieval.

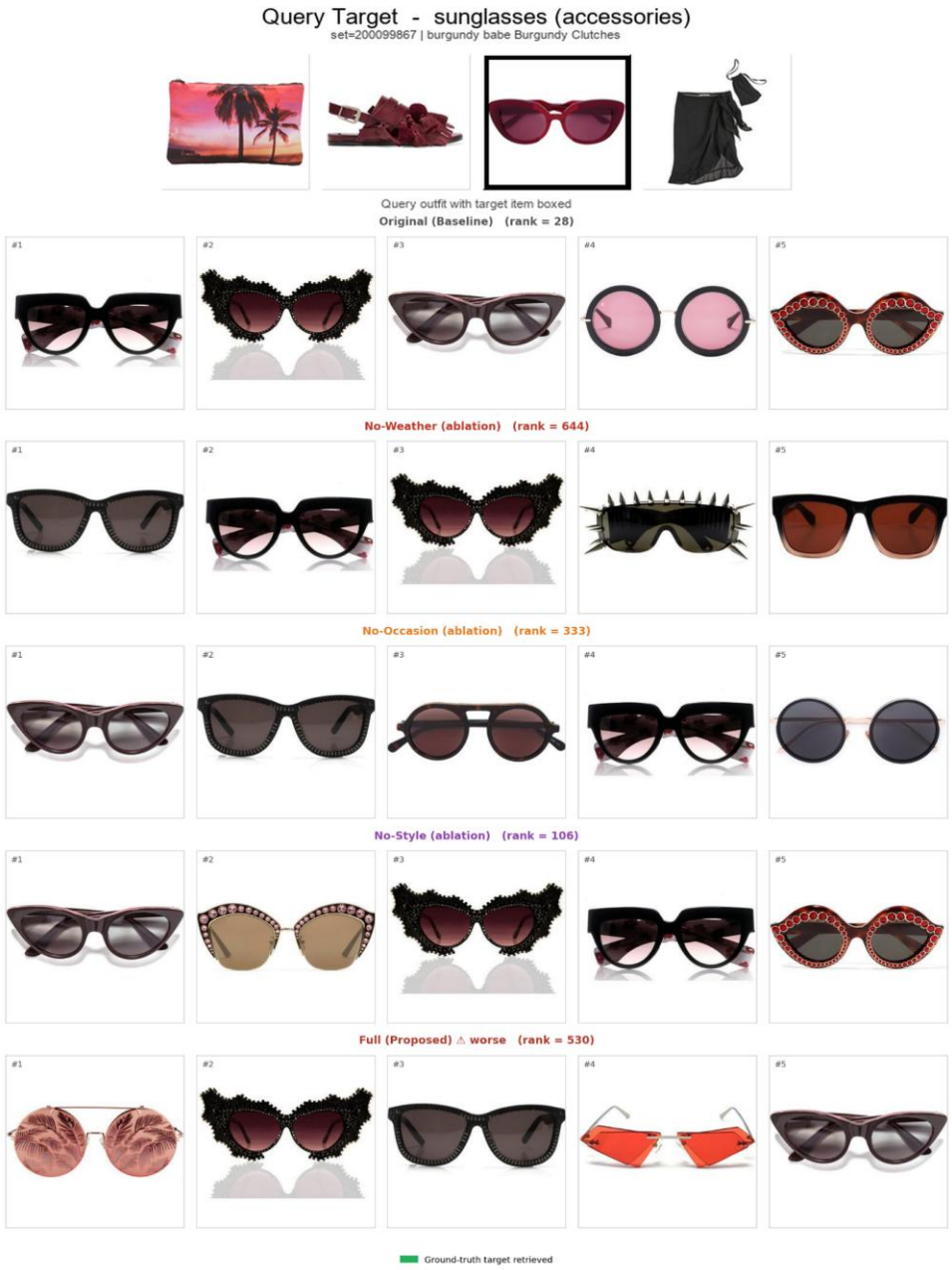


Figure A3: Representative failure case for a sunglasses target item. In this case, the original text condition ranks the target sunglasses higher than the context-aware description. The context-aware description shifts the retrieval direction toward broader sunset or beach-related visual cues, while failing to preserve the fine-grained color and shape attributes needed for the sunglasses target.